

STA 2100: PROBABILITY AND STATISTICS I ASSIGNMENT

QUESTION 1

(a) A national health ministry intends to survey the Body Mass Index (BMI) of citizens. A total of 24,000 individuals are distributed across six provinces as shown in the table below.

Province	Nairobi	Central	Coast	Rift Valley	Eastern	Western
Population	6 000	5 000	4 000	4 500	2 500	2 000

A stratified sample of 480 individuals is to be selected using proportional allocation.

- (i) Determine the sampling fraction and calculate the number of individuals to be selected from each province.
- (ii) Describe in detail how you would carry out *systematic sampling* to obtain the required sample from **Nairobi Province**. State the sampling interval k .
- (iii) State **two** advantages and **one** disadvantage of probability sampling over non-probability sampling.
- (iv) Distinguish between *cluster sampling* and *multi-stage sampling*, illustrating each with one practical example.

(b) The ages (in completed years) of 40 patients who attended an outpatient clinic on a given day are listed below.

23	45	31	67	28	52	41	35	19	58
44	37	62	27	48	33	71	25	54	39
43	29	65	38	21	57	32	46	72	26
51	36	68	30	42	22	59	34	47	60

- (i) Using the rule $2^k > n$ to determine the number of classes k , and clearly showing **all** steps (range, class width, class limits, class boundaries, tally), construct a grouped frequency distribution table that also includes a cumulative frequency column.
- (ii) Using the grouped frequency distribution in (i), draw a *histogram* and superimpose a *frequency polygon* on the same set of axes. Fully label your diagram.
- (iii) Construct a *stem-and-leaf diagram*, using the tens digit as the stem. Arrange the leaves in ascending order and briefly comment on the shape of the distribution.

(c) Describe any **four** methods that can be used to collect primary data in a statistical survey. For each method, state its name and briefly explain how it is carried out.

QUESTION 2

The daily calorie intake (in hundreds of kilocalories, 00kcal) of 80 adults participating in a nutrition study is recorded in the frequency distribution below.

Calorie intake	$10 < x < 14$	$14 < x < 18$	$18 < x < 22$	$22 < x < 26$	$26 < x < 30$	$30 < x < 34$
No. of adults	5	12	22	21	15	5

(a) Estimate the following measures of **central tendency and relative position**:

- (i) The arithmetic mean $\bar{x}$, the median, and the modal calorie intake.
- (ii) The 5th decile D_5 and the 85th percentile P_{85} .

(b) Estimate the following measures of **dispersion**:

- (i) The variance s^2 and standard deviation s .
- (ii) The first quartile Q_1 , the third quartile Q_3 , the inter-quartile range (IQR), and the semi inter-quartile range (SIQR).
- (iii) The coefficient of variation (CV). Comment on what it reveals about the spread of the distribution relative to the mean.

(c) Assess the **shape** of the distribution:

- (i) Compute **Pearson's coefficient of skewness** and **Bowley's coefficient of skewness**. Interpret both results and comment on whether they are mutually consistent.
- (ii) Compute the **moment coefficient of kurtosis** and state, with justification, whether the distribution is leptokurtic, mesokurtic, or platykurtic.

(d) A student's academic performance in five university courses is recorded below. Compute the **weighted mean grade** and interpret the result.

Course	Credit hours (weight)	Score (%)
Statistics	3	72
Mathematics	4	68
English	2	80
Computer Science	3	75
Economics	2	70

(e) The number of defective items per batch in five consecutive production runs is: 2, 4, 8, 16, 32.

- (i) Calculate the **geometric mean** of the five values.
- (ii) Calculate the **harmonic mean** of the five values, and compare it with the geometric mean and the arithmetic mean.

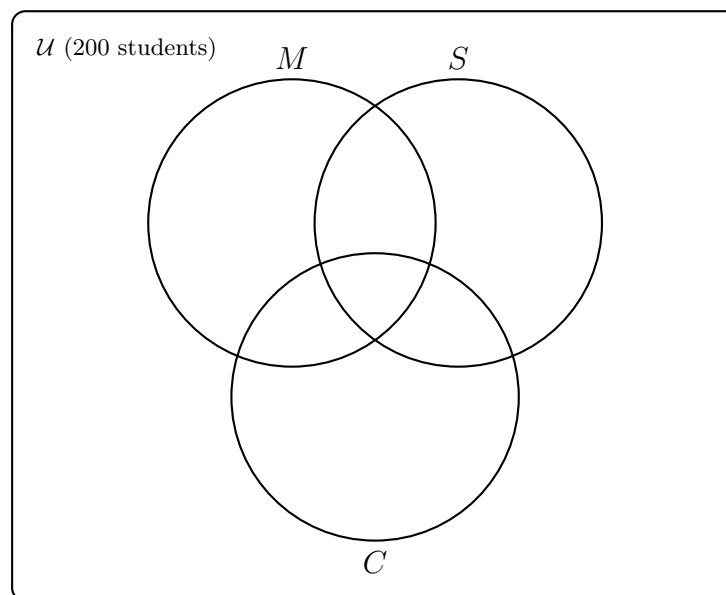
QUESTION 3

A survey of 200 university students investigated enrolment in three elective units: **Mathematics** (M), **Statistics** (S), and **Computer Science** (C). The following information was established:

- 95 students study Mathematics; 80 study Statistics; 70 study Computer Science.
- 35 students study both Mathematics and Statistics.
- 30 students study both Mathematics and Computer Science.
- 25 students study both Statistics and Computer Science.
- 15 students study all three units.

(a)

- (i) Draw a clearly labelled Venn diagram representing the universal set $\mathcal{U}$ of all 200 students. Calculate and fill in the number of students in **each of the eight regions** (the seven intersecting regions and those studying none of the three units). An empty template is provided below.



- (ii) Using your completed Venn diagram, fill in the two-way frequency table below for events M and S .

	S	S'	Total
M			
M'			
Total			200

(b) A student is selected at random from the 200. Find the probability that the student:

- (i) Studies **only** Mathematics.
- (ii) Studies **exactly two** of the three units.
- (iii) Studies **at least one** of the three units.
- (iv) Studies **none** of the three units.
- (v) Studies Mathematics **or** Statistics, **but not both**.

(c)

- (i) Given that the selected student studies Computer Science, find the probability that the student **also** studies Statistics.
- (ii) Given that a student studies **at least one** of the three units, find the probability that they study **all three**.

(d)

- (i) Test whether events M and S are *statistically independent*. Show all calculations and state your conclusion clearly.
- (ii) Using the *inclusion–exclusion principle*, compute $P(M \cup S \cup C)$ and verify that your answer is consistent with the Venn diagram.

(e) Three students are selected at random *without replacement* from the 200. Find the probability that **all three** study at least one of the three units.

QUESTION 4

(a) Bivariate Data

The table below shows the weekly number of hours studied (x) and the corresponding end-of-semester test score (y) for eight students.

Student	1	2	3	4	5	6	7	8
Hours, x	2	3	5	6	7	8	9	10
Score, y	55	60	68	72	75	80	85	90

Use your calculator to provide the following sums: n , $\sum x$, $\sum y$, $\sum x^2$, $\sum y^2$, $\sum xy$

- Compute Pearson's product moment correlation coefficient r and comment on the nature and strength of the linear relationship between x and y .
- Determine the least-squares regression line $\hat{y} = a + bx$, and use it to estimate the test score of a student who studies 11 hours per week.
- Assign ranks to x and y , compute Spearman's rank correlation coefficient and compare the result with r obtained in (i).

(b) Discrete Random Variables

A discrete random variable X has the probability mass function shown below.

x	-1	0	1	2	3
$P(X = x)$	0.10	k	0.25	0.30	0.15

- Find the value of k and verify that $P(X = x)$ constitutes a valid probability mass function.
- Calculate $E(X)$, $E(X^2)$, and $\text{Var}(X)$.
- $Y = 3X - 4$, determine $E(Y)$ and $\text{Var}(Y)$ without constructing the probability distribution of Y .
- Find $P(-1 \leq X \leq 2)$ and $P(X > 1)$.

(c) The Binomial Distribution

A diagnostic test for a particular disease has a 70% success rate in correctly identifying positive cases. A random sample of 10 patients known to have the disease are tested independently. Let X denote the number of patients whose disease is correctly identified.

- Find the probability that **exactly** 7 patients are correctly identified.
- Find the probability that **at least** 8 patients are correctly identified.
- Find the probability that **fewer than** 3 patients are correctly identified.
- Determine the **expected number** of correct identifications and the **standard deviation**.